

UNIT 41 ENGINEERING REPORT.

Manufacturing Secondary Machining Processes.

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INTRODUCTION.

Background and Scope.

This report provides a detailed account of the safe setup, practical operation, and systematic optimisation of two traditional secondary machining processes. The study involved the manufacture of two distinct engineering components:

1. A plumb bob, which is a symmetrical, pointed weight used in construction and surveying to establish a vertical reference line, was manufactured from aluminium using a centre lathe.
2. A rectangular carbon steel bar with precision-drilled holes was produced using a vertical milling machine.

Secondary machining processes are essential in general engineering, maintenance, and bespoke component production. Although computer numerical control (CNC) machining is widespread, manual turning and milling skills remain critical for prototype development, one-off repairs, toolroom operations, and scenarios where the setup time for automated production is not economically justified. This assignment focuses on manually operated machines to enhance understanding of cutting mechanics, tooling, workpiece interaction, and the direct cause-and-effect relationships that influence machining. The two components were selected for their contrasting characteristics. The aluminium plumb bob requires tapered turning, facing, parting, and potentially knurling or drilling for a suspension loop. These operations are sensitive to spindle speed, feed rate, tool geometry, and workholding rigidity. In contrast, the rectangular carbon steel bar requires precise work holding, edge finding, centre drilling, and through-hole drilling at defined coordinates, with particular attention to drill wander, burr formation, and cutting fluid application.

Purpose and Learning Aims.

The primary objective of this assignment is to demonstrate and provide evidence of competence in four interconnected areas of manual machining:

- Safe machine setup, including workpiece and tool mounting, guard positioning, speed and feed selection, and pre-operational checks.
- Risk assessment and hazard control, identifying foreseeable hazards (entanglement, ejected work, sharp swarf, tool breakage, hot chips) and implementing proportional controls.
- Accurate machining techniques, achieving dimensional tolerances, geometric form (concentricity, flatness, perpendicularity), and specified surface finishes.
- Measurement and quality control, using appropriate metrology (micrometres, vernier callipers, depth gauges, dial test indicators) to check compliance with a given engineering drawing.

In addition to these technical aims, the report addresses the ongoing refinement of machining processes. Observations made during production were used to adjust parameters, tooling, or setups to improve outcomes in real time. This iterative approach reflects industrial practice, where initial components are inspected and processes are modified before commencing full production.

Legislative and Regulatory Context.

All machining activities described in this report were conducted with reference to relevant health and safety legislation applicable within the United Kingdom. The principal statutes and regulations that inform safe working practices include:

- Health and Safety at Work etc. Act 1974 (HSWA): the overarching legal framework placing duties on employers plus employees to ensure health and safety. Under Section 7, the operator has a duty to take reasonable care of their own safety and that of others who may be affected by their acts or omissions.
- Management of Health and Safety at Work Regulations 1999: requiring systematic risk assessment and, where necessary, implementation of preventive and safeguarding measures.

- Provision and Use of Work Equipment Regulations 1998 (PUWER): mandating that work equipment (including lathes and milling machines) is suitable for the purpose, properly maintained, and fitted with adequate guards and emergency stops.
- Personal Protective Equipment at Work Regulations 1992 (as amended): requiring provision and use of appropriate PPE, including safety spectacles (or a full face visor), hearing protection (where noise exposure exceeds action values), and safety footwear.

These legal requirements were implemented through local workshop rules, machine-specific safe operating procedures, and dynamic risk assessments conducted during each machining task.

Structure of the Report.

Following this introduction, the report is organised into five main sections:

1. Risk Assessment and Hazard Control: a tabulated risk assessment covering both lathe and milling operations, with identified hazards, likelihood/severity ratings, and control measures (e.g, guarding, swarf hooks, brush not fingers, chipbreaking tool geometry, eye protection).
2. Component 1: Manufacture of the Aluminium Plumb Bob on a Centre Lathe: covering material selection (e.g, EN 6082 or 6061 aluminium), work holding (three-jaw chuck or collet), tooling (HSS or carbide tipping), step-by-step operations (facing, turning the taper, drilling for a cord, knurling if required, parting off), speeds and feeds, and quality checks including diameter, concentricity, and surface finish.
3. Component 2: Manufacture of the Rectangular Carbon Steel Bar with Drilled Holes on a Vertical Milling Machine: including work holding (machine vice with parallels), edge finding (using an edge finder or wobbler), datum setting, centre drilling to prevent drill wander, drilling to final size (e.g, using a twist drill in a drill chuck), deburring, and inspection of hole positions and diameters.
4. Measurement and Quality Control: a comparative analysis of specified tolerances versus achieved measurements, using calibrated instruments, and a discussion of measurement uncertainty.
5. Continuous Refinement of Machining Processes: a log of process adjustments made during manufacture, such as modifying cutting speeds to reduce built-up edge on aluminium, altering feed pressure to avoid taper

chatter, or adjusting clamping to prevent workpiece lifting during drilling on the mill. This section demonstrates iterative learning and practical problem-solving.

Expected Outcomes.

By the conclusion of this report, the reader will have a clear, evidence-based understanding of how two traditional secondary machining processes can be executed safely, accurately, and efficiently. More importantly, the report will demonstrate that competence in manual turning and milling is not merely the ability to produce a component to a drawing, but rather the integration of hazard awareness, machine sympathy, measurement discipline, and reflective practice. These skills are directly transferable to more advanced machining contexts, including CNC programming and multi-axis operations.

The following sections provide a detailed, operation-by-operation account of how both components were produced, accompanied by annotated photographs, dimensional inspection reports, and a reflective commentary on the refinements made during the process.

Engineering Context and Machining Theory.

Definition and Role of Secondary Machining Processes.

Secondary machining processes refer to material removal operations carried out on a workpiece after primary forming processes (such as casting, forging, extrusion, or rolling). While primary processes define the general shape and grain structure of a material, they rarely achieve the dimensional accuracy, surface finish, or geometric features (holes, threads, tapers, slots) required in an engineering drawing. Secondary machining bridges this gap.

Common secondary processes include turning, milling, drilling, facing, boring, reaming, tapping, and grinding. This report focuses on three core operations: turning (producing the conical plumb bob), milling (producing the rectangular bar's flat

surfaces and accurate datum edges), and drilling (producing holes in the carbon steel bar).

Turning on a Centre Lathe.

Principle of Operation.

In turning, the workpiece is rotated about its axis while a stationary, single-point cutting tool traverses along or across the workpiece. The centre lathe achieves this through a spindle that rotates the workpiece (held in a chuck or between centres), a tailstock that provides support or holds drilling tools, and a carriage that moves the tool in longitudinal (Z-axis) or transverse (X-axis) directions.

Key sub-operations used in manufacturing the aluminium plumb bob:

- Facing: The tool moves radially across the end of the workpiece to create a flat, smooth surface square to the axis.
- Straight turning (parallel turning): Reduces diameter over a length.
- Taper turning: The tool moves simultaneously in longitudinal and transverse directions (using the compound slide or offset tailstock) to produce a conical form.
- Parting (cut-off): A narrow, blade-like tool cuts radially through the workpiece to separate the finished component.
- Drilling (on lathe): A drill bit held in the tailstock is fed into the rotating workpiece to create a centred hole for the suspension cord.

Cutting Mechanics and Chip Formation.

Material removal in turning occurs by shear deformation. The cutting tool's rake angle, relief angle, and cutting edge geometry determine how the chip flows, how much heat is generated, and the quality of the machined surface.

For aluminium (e.g, 6082 or 6061), which is relatively soft and ductile but prone to built-up edge (BUE) formation, the following theoretical considerations apply:

- High cutting speeds (typically 200-600 m/min for uncoated carbide) promote shear-localised chip formation and reduce BUE.
- Sharp, polished tools with positive rake angles (10° - 20°) minimise friction and cutting forces.
- Generous cutting fluid (or mist lubrication) improves surface finish and prevents aluminium from adhering to the tool tip.

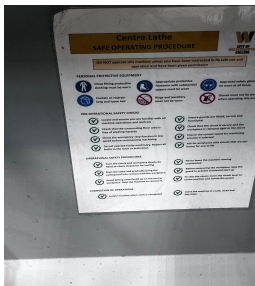
For carbon steel, which has higher strength and work-hardening potential, lower cutting speeds (30-80 m/min for HSS tools, 100-250 m/min for carbide) and robust tool geometries (zero or negative rake for interrupted cuts) are typical.

Speed, Feed, and Depth of Cut Relationships.

The three fundamental cutting parameters are interdependent:

- Spindle speed N (rpm) is calculated from desired cutting speed and workpiece diameter D (mm).
- Feed rate f (mm/rev) determines the distance the tool advances per revolution. For finishing, low feeds (0.05-0.15 mm/rev) are used; for roughing, higher feeds (0.2-0.5 mm/rev).
- Depth of cut a (mm) is the radial thickness removed per pass. Light finishing cuts (0.1-0.5 mm) follow roughing cuts (1-3 mm) to achieve final dimensions.

Incorrect selection leads to chatter, poor surface finish, tool breakage, or machine overload.



(fig.11 safety rules for operating the lathe)

Milling and Drilling on a Vertical Milling Machine.

Principle of Operation.

Unlike turning, the vertical milling machine rotates the cutting tool (an end mill, face mill, or drill bit) while the workpiece remains clamped to a movable table. The spindle holds the tool; the table moves in X (left-right), Y (forward-back), and sometimes Z (up-down) axes.

For the rectangular carbon steel bar, the primary operations were:

- Peripheral milling (slab milling): Produces flat surfaces parallel to the cutter axis - used to square the bar.
- Drilling: A twist drill is fed vertically into the workpiece to generate through holes or blind holes.

- Work holding: The bar is clamped in a machine vice, which itself is aligned to the table using a dial test indicator.

Drilling Theory and Hole Quality.

Drilling in carbon steel presents several challenges:

- Drill wander: Flexible long drills or misaligned centres cause holes to deviate from true position. Prevention requires a centre drill (spotting drill) to create a starter dimple before the final twist drill.
- Chip evacuation: Carbon steel produces long, stringy chips that can wrap around the drill. Peck drilling (repeatedly retracting the drill) breaks chips.
- Cutting fluid: Soluble oil or cutting paste is essential to reduce friction and prevent work hardening at the hole wall.
- Burr formation: On the exit side of a through hole, a raised burr is inevitable. This is removed by a deburring tool or larger drill chamfer.

Hole positional accuracy depends on proper edge finding (using a wobbler or electronic edge finder) and understanding backlash in the handwheels. All movements should be made in the same direction to eliminate backlash error.

Hazard Analysis in Machining Theory.

At a theoretical level, machining hazards arise from the conversion of electrical energy into mechanical work and heat. Understanding why hazards exist is essential for risk control.

Rotational Hazards (Entanglement).

Both lathes (rotating workpiece) and milling machines (rotating tool) present entanglement risks. Loose clothing, long hair, gloves, or jewellery can be caught and wrapped around the rotating member, drawing the operator into the machine. This mechanism is rapid and causes severe crush, avulsion, or fracture injuries. Theoretical basis: angular momentum and friction between the rotating surface and fibrous material produce a winding force that is almost impossible to resist once initiated.

Flying Debris and Chip Projection.

At cutting speeds of 200-1000 m/min, chips are ejected with high kinetic energy. Small, hot chips can travel several metres. In milling, broken cutter teeth or entire end mills can be flung from the spindle. Risk factors: lack of guarding, improper tool clamping, or use of damaged tooling.

Tool Breakage and Workpiece Ejection.

Excessive cutting forces (due to too high feed rate, too deep a cut, or blunt tool) can cause the tool to fracture. Fragments become high-velocity projectiles. Similarly, inadequate work holding, such as an incorrectly tightened chuck jaw or a machine vice not clamped to the table, can lead to the workpiece being torn from its fixture and ejected.

Heat Generation and Thermal Hazards.

Machining converts almost all mechanical energy into heat. For example, drilling carbon steel at 1000 rpm with moderate feed can generate workpiece surface temperatures exceeding 200°C. Chips are often blue-straw coloured from oxidation, indicating temperatures above 300°C. Contact with hot chips or a hot workpiece causes burns. Prolonged contact with the machine's spindle housing (if inadequately cooled) can also cause discomfort.

Sharp Edges and Swarf.

Machined edges, drilled hole perimeters, and especially lathe-produced swarf (continuous, razor-sharp ribbons) are capable of causing deep lacerations. Swarf can also wrap around rotating chucks and become a whipping hazard. Control: Never clear swarf by hand while the machine is rotating; use a swarf hook or brush, and stop the machine for tangled swarf.

Theoretical Justification for Control Measures.

The hierarchy of controls (Elimination → Substitution → Engineering controls → Administrative controls → PPE) is applied to machining hazards:

Hazard	Engineering control	Administrative control	PPE
Entanglement	Fixed and interlocked guards, chuck guards, emergency stop.	No gloves, tied hair, no jewellery.	Fitted overalls (not loose).

Flying debris	Transparent chip guard (eye shield).	Mandatory safety briefing.	Safety spectacles (side shields) or full-face visor.
Tool breakage	Correct speed/feed selection, tool overhang minimised.	Pre-use tool inspection.	Face shield for high-risk ops.
Heat & burns	Coolant; chip tray to contain hot chips.	Wait before touching work.	Heat-resistant gloves (only when the machine is stationary).
Sharp swarf	Chip breaker tool geometry.	Never clear by hand.	Leather-reinforced gloves for housekeeping.

Relevance to the Two Manufactured Components.

- Aluminium plumb bob (lathe): The low cutting forces and high surface speed requirement make it prone to BUE(Built-Up Edge) . Theory suggests increasing speed and using lubricant. Entanglement risk is significant due to rotating chuck jaws; therefore, no rings or loose sleeves. The taper turning operation requires precise compound slide angle setting; a small angular error (0.1°) leads to measurable diameter deviation over the taper length.
- Carbon steel bar (mill): Higher cutting forces demand rigid work holding. Drilling theory indicates that without a centre drill, holes will wander, especially in harder materials. Burr formation is inevitable; the report will show deburring as a separate quality control step. Climb milling versus conventional milling theory must be understood, for manual milling, conventional milling is generally safer because it eliminates the risk of the cutter pulling the workpiece into the tool.

Summary of Engineering Context.

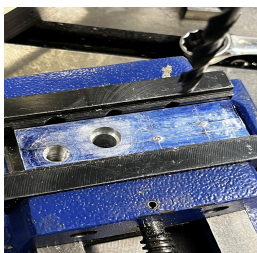
This section has established the theoretical foundation for safe and effective secondary machining. Key takeaways include:

- Turning and milling differ fundamentally in whether the workpiece or tool rotates.
- Cutting parameters (speed, feed, depth of cut) are derived from material properties and tooling.
- Each hazard (entanglement, debris, breakage, heat, sharp edges) has a predictable physical cause.
- Control measures are not arbitrary but are grounded in the laws of mechanics and heat transfer.
- The two components, plumb bob and rectangular bar, present distinct machining challenges that will be addressed in subsequent sections by applying this theory to practical setups.

The following sections will demonstrate how this theory was translated into step-by-step machine setup, operation, and refinement, with full attention to the legal and practical requirements established in the introduction.

Vertical Drilling Machine (Pillar Drill).

Used for: Rapid hole production and countersinking in the carbon steel bar.



(fig.17 my workpiece in progress of machining)

Step 1: Secure Workpiece in Machine Vice

- Action: Place the bar in a drill vice. Ensure it is supported by parallels and the vice is bolted to the drill table.
- Rationale: Drilling generates high torque. If the work is not bolted down, the drill can "grab" the metal and spin it at high speed (the "helicopter effect"), causing severe injury.

Step 2: Set Spindle Speed (Belt or Gear)

- Action: Adjust the machine belts or gears to a lower speed for steel (e.g., 600-900 rpm for an 8mm bit).
- Rationale: Steel is a poor conductor of heat. Excess speed will "burn" the drill's cutting edge, making it blunt and potentially work-hardening the hole.

Step 3: Align and Adjust the Telescopic Guard

- Action: Pull the spring-loaded guard down to cover the chuck and drill bit.
- Rationale: Complies with PUWER Regulation 11. It prevents entanglement with the rotating spindle and contains flying swarf.

Step 4: Manual Feed with "Peck" Action

- Action: Use the hand lever to feed the drill. Apply firm pressure, then retract slightly every 2-3mm.
- Rationale: This breaks the "stringy" carbon steel swarf into smaller, safer chips and allows cutting fluid to reach the drill tip.



(fig. 16 drills i will use on my workpiece)

Step 5: Final Breakthrough Control

- Action: Ease off the pressure as the drill starts to exit the bottom of the bar.
- Rationale: This is the most dangerous moment where the drill is likely to "snag" and try to tear the workpiece out of the vice.



(fig.15 setting up my workpiece on the pillar drill)

Core Hazards and Safety Controls.

Rotating Spindle and Chuck.

The primary risk is entanglement. To control this, operators must never wear gloves, ensure long hair is tied back, and keep sleeves tight or rolled up. Additionally, a telescopic chuck guard must be used to provide a physical barrier between the operator and the rotating parts.

Spinning Workpiece.

The risk is impact or severe laceration. Because a drill bit can "grab" the metal as it cuts, the workpiece must never be held by hand. The only safe control is to securely bolt the machine vice to the drill table using T-bolts to prevent the part from spinning out of control.

Projected Hot Swarf.

The risks are burns and eye injuries. As the drill removes material, hot, sharp metal chips are ejected at speed. Control measures include wearing safety spectacles or a full-face visor and applying cutting fluid to keep the swarf cool and manageable.

Alignment with Safe Working Practices.

These steps are not just shop rules; they follow specific UK health and safety laws to ensure a Safe System of Work:

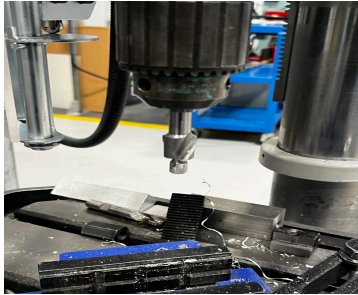
- PUWER 1998 (Provision and Use of Work Equipment Regulations): Using the telescopic guard and bolting down the vice directly follows Regulation 11, which mandates that dangerous moving parts are guarded and workpieces are stable to prevent ejection.
- Health and Safety at Work Act 1974 (Section 7): By removing gloves and tying back hair, the operator is fulfilling their legal duty to take "reasonable care" of their own safety and follow the employer's safety instructions.
- MHSWR 1999 (Management Regulations): Applying the "Peck Drilling" technique is an administrative control resulting from a risk assessment. It prevents the build-up of long, stringy swarf, which reduces the risk of entanglement and fire from overheating.
- PPE Regulations 1992: Wearing eye protection serves as the "last line of defence" required by law when engineering controls (like the guard) cannot stop every single flying chip.

Conclusion: Selection of the Vertical Drilling Machine.

The vertical drilling machine is the most efficient choice for producing holes in the carbon steel bar because its specialized design focuses entirely on axial rigidity. While a milling machine is more versatile, the pillar drill allows for rapid, repeatable hole production with a direct "feel" for the cutting pressure through the manual feed lever. This feedback is critical when drilling tougher materials like carbon steel, as it allows the operator to detect "grabbing" or tool dulling immediately.

By using a dedicated drilling setup, the workshop maintains a clear Safe System of Work, ensuring that high-torque operations are contained within a machine

specifically built to handle the vertical forces and swarf evacuation requirements of hole-making.



(fig 18. Drill bit secured safely in the the chuck)

Health and Safety Regulations (Detailed Analysis).

Machining operations, turning and milling, involve rotating machinery, sharp tools, ejected material, and high temperatures. Compliance with health and safety legislation is not merely a bureaucratic requirement; it is the foundation of safe systems of work. This section provides a detailed analysis of four key regulations, explaining their specific requirements and demonstrating how they were applied to the manufacture of the aluminium plumb bob and the rectangular carbon steel bar.

Management of Health and Safety at Work Regulations 1999.

The Management Regulations (often abbreviated to MHSWR) impose a duty on employers (and, in educational or workshop settings, on those in control of premises) to systematically manage risk. The core requirements are:

- Identification of hazards: anything with the potential to cause harm.
- Risk assessment: evaluating the likelihood and severity of harm.
- Implementation of control measures: putting in place proportionate actions to reduce risk.
- Review: updating the assessment when circumstances change.

A risk assessment must be suitable and sufficient. This means it must:

- Cover all significant hazards (not trivial ones).
- Be specific to the task, not generic.
- Be based on reliable information (e.g. manufacturer's data for machine speeds).

- Remain valid; if new equipment or a different material is introduced, the assessment must be reviewed.

Application to this assignment.

Before any metal was cut, a detailed risk assessment was completed for both the centre lathe and the vertical milling machine. The following significant hazards were identified:

- Rotating chuck (lathe): entanglement, ejection of work if loose.
- Rotating milling cutter: entanglement, ejection of broken tool fragments.
- Sharp swarf: lacerations (continuous ribbon from turning aluminium, stringy chips from drilling steel).
- Hot swarf and hot workpiece: burns (blue chips from carbon steel drilling).
- Flying debris: chips, broken tool tips.
- Tool breakage: sudden release of energy.
- Workpiece ejection: inadequate clamping.

For each hazard, control measures were implemented before machining began. For example:

Hazard	Control measure (pre-task).	Regulation reference.
Rotating chuck	Chuck guard closed; no gloves; hair tied back; sleeves short or tight.	MHSWR 1999, PUWER 1998.
Hot swarf from drilling steel	Cutting fluid applied; wait 10 seconds after spindle stop before touching the workpiece.	MHSWR 1999.
Flying chips	Full-face visor or safety spectacles with side shields.	MHSWR, PPE Regs 1992.

Crucially, the risk assessment was continuously reviewed during machining. When drilling the carbon steel bar on the mill, it was observed that a long, continuous swarf was wrapping around the drill. This created an unexpected entanglement hazard. The assessment was immediately updated, and the control measure changed from “peck drill every 5 mm” to “peck drill every 2 mm and clear swarf with a brass hook after each retraction”. This dynamic review is a legal requirement under Regulation 3(1) of MHSWR, which states that assessments must be “suitable and sufficient” and that “any significant change” triggers a review.

Personal Protective Equipment Regulations 1992

The PPE Regulations 1992 (as amended) require that suitable personal protective equipment is provided (by the employer) and used (by the employee) where risks cannot be adequately controlled by other means. PPE is always the last line of defence in the hierarchy of controls (after elimination, substitution, engineering controls, and administrative controls). It does not reduce the hazard; it merely reduces the consequence if other controls fail.

Required PPE for machining.

Based on the risk assessment:

- Safety spectacles with side shields (or full-face visor): Protects eyes from flying chips, coolant spray, and broken tool fragments. A visor is preferred for high-risk operations such as parting off on the lathe or drilling deep holes in steel.
- Steel-toe capped boots: Protects feet if a heavy workpiece (e.g, the carbon steel bar, or a chuck key dropped) is dropped. Also provides slip resistance on oily workshop floors.
- Overalls (close-fitting, no loose drawstrings): Prevents entanglement by covering loose personal clothing. Cotton or flame-resistant material is preferred to avoid melting if hot chips land on fabric.
- Hearing protection (earplugs or earmuffs): Required if noise levels exceed 80 dB(A) (action value). Machining steel with carbide tools can easily reach 90–95 dB(A) during interrupted cuts or milling.

Why PPE is essential but not sufficient:

Both turning and milling produce sharp, hot swarf. A lathe cutting aluminium generates a long, razor-sharp ribbon that can whip around and cut exposed skin. A milling cutter throwing a hot steel chip can cause corneal abrasion or a burn. PPE directly prevents these injuries. However, PPE does not prevent entanglement, only guarding and administrative controls (no gloves) can do that. Therefore, PPE must always be used alongside the hierarchy.

Application to this assignment

- Aluminium plumb bob (lathe): Safety glasses worn at all times. Face visor used during parting off, as this operation generates the most flying fragments. Overalls tucked in; no ring or watch.
- Carbon steel bar (mill): Safety glasses plus additional face shield when drilling, because drill breakage is more likely in steel. Hearing protection worn during milling passes (end mill cutting steel is loud). Steel-toe boots worn throughout, as the bar weighed approximately 2 kg and could cause injury if dropped.

Provision and Use of Work Equipment Regulations 1998 (PUWER).

PUWER is arguably the most critical regulation for machining environments. It applies to all work equipment, from hand tools to CNC machines, and requires that equipment is:

- Suitable for the intended task: e.g, a centre lathe used for turning aluminium must have appropriate speed range (not just a wood lathe).
- Safe for use: guards, interlocks, emergency stops are functional.
- Properly maintained: cleaning, lubrication, inspection schedule.
- Used only by trained persons: competence evidence required.
- Accompanied by safe systems of work: written procedures, warning signs.

Specific PUWER requirements relevant to this assignment:

- Regulation 11 (dangerous parts of machinery): All rotating parts (chuck, chuck key, drive belt, milling spindle, drawbar) must be guarded. The lathe chuck guard and milling cutter guard must be interlocked (or at least fixed in position).
- Regulation 19 (markings): Machine must have clear emergency stop buttons, direction arrows, and speed selector markings.
- Regulation 22 (controls): Start and stop controls must be easily accessible. The lathe's emergency stop footbrake (if fitted) must be tested before use.
- Regulation 25 (training): No person may use work equipment unless they have received adequate training. This assignment was carried out only after passing a machine-specific competency test.

Application to this assignment.

- Pre-use inspection: Before turning the plumb bob, the lathe chuck jaws were inspected for wear and securely tightened. The chuck key was removed

immediately after use (a legal requirement under PUWER to prevent ejection). The chuck guard was closed and the interlock verified.

- Guards in place: On the mill, the cutter guard (transparent polycarbonate) was lowered before spindle rotation. The guard is not interlocked on the particular machine, but a written safe system of work requires verbal confirmation (“guard down”) before pressing start.
- Emergency stops checked: Both machines have prominent red mushroom-head emergency stops. Each was tested at the beginning of the session: press stop, machine powers down; twist reset, machine can be restarted. This takes 5 seconds and is legally required under PUWER Regulation 30(1) (systems to stop equipment).
- Only trained operation: The operator (the author) had completed a one-day induction covering speeds/feeds, tool mounting, work holding, and emergency procedures. No untrained persons were allowed within the working envelope.

Consequences of non-compliance with PUWER:

Without proper controls, a loose chuck key becomes a projectile (kinetic energy of a 0.5 kg key at 500 rpm $\approx$ 340 J equivalent to a small hammer throw). Workpiece ejection from an incorrectly tightened chuck can cause fatal head injuries. PUWER is the primary regulation that mandates specific physical and administrative controls to prevent these disasters.

Health and Safety at Work Act 1974 (HSWA).

The HSWA is the enabling statute under which all other UK health and safety regulations are made. It places duties on both employers and employees. For this assignment, the most relevant sections are:

- Section 7 (General duties of employees at work): Every employee must take reasonable care for their own health and safety and for others who may be affected by their acts or omissions. They must also cooperate with their employer on health and safety matters.
- Section 8 (Duty not to interfere with or misuse anything provided for safety): It is an offence to deliberately remove a guard, disable an interlock, or bypass an emergency stop.

Application to this assignment.

As the operator (acting in the role of an employee or trainee engineer), the following specific actions were taken to comply with Section 7:

- Work safely: The correct speeds and feeds were selected (from Section 2 theory). A test cut was taken before final dimensions. The machine was stopped before measuring or clearing the swarf.
- Follow instructions: Workshop rules (displayed on the wall) were followed: no loose clothing, no compressed air to clean machines, use a brush not fingers for swarf.
- Protect others: The area around the lathe and mill was cordoned off with warning signage. Other workshop users were asked to stand behind the safety line during high-risk operations (parting off, deep drilling). Swarf was cleared immediately after each operation to prevent slip hazards.
- Do not misuse safety equipment: The chuck guard was never defeated. The milling cutter guard was never propped open. The emergency stop was used only in genuine emergencies (not as a routine stop, which would wear out components).

Legal consequences of failure.

Failure to follow the HSWA can result in injury. For example:

- Rotating parts hazard: If an operator wore a loose-sleeved shirt while turning aluminium on the lathe, the sleeve could wrap around the rotating chuck. The resulting force would pull the operator's arm into the machine, causing spiral fracture, degloving, or amputation. This is a foreseeable consequence of non-compliance with Section 7(1).
- Flying debris hazard: If the milling cutter guard was raised to get a better view, a broken carbide end mill tooth could be ejected at high speed. At typical milling speeds (1000–3000 rpm), a 5 g fragment has kinetic energy comparable to a .22 bullet. This could cause penetrating eye injury or facial fracture. This is a direct breach of Section 8 (misuse of safety equipment).

Summary of Regulatory Application.

Regulation.	Key requirement.	Applied to lathe (plumb bob).	Applied to a mill (carbon steel bar).
MHSWR 1999	Risk assessment, review.	Assessed rotating chuck, swarf; reviewed	Assessed drill breakage, work ejection; reviewed

		when taper chatter occurred.	when swarf wrapped.
PPE Regs 1992	Last-line defence.	Safety glasses, visor (parting), overalls, boots.	Glasses, hearing protection (milling steel), boots, gloves for deburring only.
PUWER 1998	Equipment safe, trained use.	Guard closed, chuck key removed, E-stop tested.	Cutter guard lowered, vice secure, training record held.
HSWA 1974 (s.7,8)	Employee duty, no misuse.	No loose clothing; used brush not hand; followed rules.	Did not override guard; cleared swarf; protected others.

Conclusion(Section).

Health and safety regulations are not abstract legal texts; they translate directly into the moment-by-moment decisions made on a workshop floor. The Management Regulations ensured that a structured risk assessment was completed before machining began. PUWER mandated that the lathe and mill were fitted with effective guards and emergency stops, and that only trained operators used them. The PPE Regulations provided the final layer of protection for eyes, feet, and hearing. And the Health and Safety at Work Act placed clear responsibility on the operator to work safely and not to tamper with safety devices.

Throughout the manufacture of the aluminium plumb bob and the carbon steel bar, these regulations were actively applied, from pre-use inspections to dynamic risk reviews when unexpected behaviour (e.g, swarf wrapping) occurred. The result was a safe working environment where the machining processes could be carried out to the required quality standards without injury. The following sections (4 and 5) will

describe the practical execution of those processes, with continuous reference back to the regulatory controls established here.

Hazard Identification and Risk Analysis (Expanded).

Machining operations, both turning and milling, involve the controlled application of high mechanical energy to remove metal. This energy, combined with rotating components, sharp tools, and hot debris, creates multiple distinct hazards. A structured hazard identification and risk analysis was undertaken before any cutting began, following the requirements of the Management of Health and Safety at Work Regulations 1999. This section expands on each of the six hazard categories, providing specific examples from the manufacture of the aluminium plumb bob and the rectangular carbon steel bar.

Mechanical Hazards

Nature of the hazard: Mechanical hazards arise from moving parts of machinery, specifically, rotating components and points where relative motion creates shear, crush, or entanglement risks.

Specific hazards in turning (plumb bob):

- Rotating chuck and spindle: The three-jaw chuck rotates at speeds up to 1500-2000 rpm depending on workpiece diameter. Any loose item (sleeve, glove, cloth, key, long hair) contacting the chuck will be instantly wrapped around it, pulling the operator into the machine.
- Rotating workpiece: The aluminium bar itself, especially if not perfectly round, presents an irregular rotating surface that can catch clothing.
- Lead screw and feed rod: These run the length of the lathe bed and rotate during powered feeds. They create pinch points and entanglement risks, particularly for operators leaning over the machine.

Specific hazards in milling (carbon steel bar):

- Rotating milling cutter: An end mill or drill chuck rotating at 500-2000 rpm. The cutter has sharp flutes that can easily cut through clothing and skin.
- Moving table (X and Y axes): The machine vice moves rapidly under power feed (or manually with long handles). Pinch points exist between the vice and fixed machine components (column, T-slot edges).
- Drawbar (top of spindle): If not fully tightened, the collet or tool holder can become loose and fall out.

Controls applied:

- No gloves, no loose clothing, no jewellery, tied-back hair (HSWA Section 7).
- Chuck guard closed and interlocked (PUWER Regulation 11).
- Machine vice handle removed after adjustment (prevents accidental engagement).
- Operator stance positioned to allow immediate withdrawal; no leaning over rotating parts.

Real-world consequence of failure: A single loose sleeve contacting a rotating chuck can pull the operator's arm into the machine, causing spiral fractures, degloving, or amputation within 0.5 seconds.

Flying Debris and Swarf

Nature of the hazard: Metal chips (swarf) are ejected from the cutting zone at high velocity, carrying significant kinetic energy. Swarf is also sharp (often razor-like) and, immediately after cutting, extremely hot.

Lathe-specific (aluminium plumb bob):

- Aluminium produces long, continuous, ribbon-like chips. At cutting speeds of 300-600 m/min, these are flung off the tool in a helical stream.
- Chips can wrap around the rotating workpiece or chuck, then whip around, creating a secondary projection hazard.
- If chip breaking is poor (e.g. wrong feed rate or tool geometry), chips can become tangled and break unpredictably.

Milling/drilling-specific (carbon steel bar):

- Milling produces small, comma-shaped chips that scatter in all directions. Steel chips are particularly dangerous because they retain heat (blue colour indicates $>300^{\circ}\text{C}$) and have sharp edges.
- Drilling (especially deep holes) produces stringy swarf that can wrap around the drill and then be thrown off in a clump.

Controls applied:

- Full-face visor or safety spectacles with side shields (PPE Regulations 1992).
- Chip guard (transparent polycarbonate) fitted to both machines.
- Use of chip-breaker tool geometry (positive rake with a step) to break aluminium chips into manageable C-shapes.
- Peck drilling (retracting every 2 mm) to break steel chips.
- Swarf cleared only when the machine stopped, using a brush or swarf hook, never fingers.

Real-world consequence of failure: A hot, sharp steel chip entering the eye causes corneal abrasion or penetrating injury. A large aluminium ribbon wrapping around the hand can cause severe laceration requiring sutures.

Tool and Workpiece Ejection

Nature of the hazard: If work holding or tool holding is inadequate, the workpiece or tool can become a high-velocity projectile. This is one of the most lethal hazards in machining because the ejected object has concentrated kinetic energy.

Lathe-specific (plumb bob):

- Workpiece ejection: If the chuck jaws are not tightened sufficiently, or if the workpiece is not fully seated, centrifugal force at high rpm can throw it out. An aluminium bar (diameter 25 mm, length 100 mm) weighing approximately 0.13 kg can be ejected at up to 20 m/s, carrying 26 J of kinetic energy, sufficient to cause skull fracture.
- Chuck key ejection: If the chuck key is left in the chuck after tightening, starting the lathe will accelerate the key to a projectile. A 0.3 kg key at 1000 rpm has tangential speed ~25 m/s, kinetic energy ~94 J, comparable to a medium-weight hammer thrown.

Milling-specific (carbon steel bar):

- Cutter ejection: An end mill or drill bit not fully tightened in the collet or chuck can be pulled out by cutting forces. The tool becomes a projectile.
- Workpiece ejection: If the machine vice is not tightened sufficiently, or if parallels are not used correctly, the steel bar can shift or be torn from the vice by the cutter's rotational force.

Controls applied:

- Chuck key removed and placed in a dedicated holder immediately after tightening (PUWER, safe system of work).
- Chuck jaws checked with a torque wrench (or by hand with adequate force) before each workpiece change.
- Machine vice clamped to the table using bolts and T-nuts; vice handle not left overnight.
- Tool collet tightened to manufacturer's torque specification; drawbar checked.
- Test pass performed before full-depth cut to verify holding integrity.

Real-world consequence of failure: A workpiece ejected from a lathe has caused fatal head injuries in documented industrial accidents. A loose milling cutter can penetrate the body or cause severe facial lacerations.

Thermal Hazards.

Nature of the hazard: Machining converts almost all mechanical energy into heat. The cutting zone can reach several hundred degrees Celsius. Both the workpiece and the chips become hot enough to cause serious burns on contact.

Lathe-specific (aluminium):

- Although aluminium has high thermal conductivity (dissipates heat quickly), the chips themselves are very hot immediately after cutting (typically 150–250°C).
- The workpiece can become warm to hot after prolonged turning, especially with high speeds and no coolant.

Milling/drilling-specific (carbon steel):

- Carbon steel has lower thermal conductivity than aluminium, so heat concentrates at the cutting edge and in the chip.
- Drilling steel without adequate coolant produces blue chips (300-400°C). The drill bit and the workpiece near the hole can reach temperatures that cause second-degree burns on contact.
- The milling cutter body can become hot after several passes.

Controls applied:

- Cutting fluid applied (soluble oil for steel; paraffin or mist for aluminium, reduces heat generation through lubrication and carries heat away.
- Wait at least 10-15 seconds after the spindle stops before touching the workpiece or tool.
- Use a temperature-indicating label (optional) or simple hand proximity test (hover hand near, not on, surface).
- Hot chips collected in a chip tray; do not allow accumulation on the floor.
- Signs posted: “Hot parts, allow to cool before handling.”

Real-world consequence of failure: Grasping a freshly drilled steel bar can cause immediate blistering burns. Swarf landing on exposed skin (forearm, neck) causes small but painful burns.

Chemical Hazards.

Nature of the hazard: Cutting fluids, lubricants, and cleaning solvents used in machining present chemical risks. Although less immediate than mechanical hazards, chronic exposure can cause dermatitis, respiratory irritation, or more serious health effects.

Specific hazards in this assignment:

- Soluble cutting oil (for steel drilling on mill): Contains emulsifiers, biocides, and extreme pressure additives. Prolonged skin contact can cause defatting, leading to irritant contact dermatitis. Mist inhalation can irritate the respiratory tract.
- Paraffin (for aluminium turning): A light mineral oil. Can cause skin dryness and irritation. Flammable, requires careful storage and no sources of ignition nearby.
- Degreasing solvents (e.g, white spirit, isopropyl alcohol): Used for cleaning components before measurement. Inhalation of vapours can cause dizziness; skin contact causes defatting.

Controls applied:

- Cutting fluid used as a fine mist or flood; local exhaust ventilation (LEV) or workshop general ventilation kept on.
- Barrier cream applied before work; hands washed thoroughly after work.
- Nitrile gloves worn only when handling chemicals (e.g, cleaning), never when the machine is rotating (gloves would increase entanglement risk).
- Solvents used in a well-ventilated area; lids replaced immediately.
- Material Safety Data Sheets (MSDS) consulted before use.

Real-world consequence of failure: Repeated exposure to cutting oil without gloves leads to dermatitis, cracked, painful skin that takes weeks to heal and makes the operator more susceptible to infection.

Noise and Vibration.

Nature of the hazard: Machining generates noise that can cause permanent hearing damage over time. Vibration affects both operator comfort and machining accuracy.

Lathe noise and vibration:

- Turning aluminium at high speed (1500+ rpm) with a sharp tool is relatively quiet (75-80 dB(A)).
- Turning steel (if applicable) or parting off can be louder (85-90 dB(A)), especially if chatter occurs.
- Vibration (chatter) occurs when the tool, workpiece, or machine stiffness is inadequate. It leaves a poor surface finish and can lead to tool breakage.

Milling/drilling noise and vibration (carbon steel bar):

- Milling steel with an end mill generates significant noise, typically 90-95 dB(A), especially during heavier cuts (2 mm depth, 0.1 mm/rev feed).

- Drilling steel produces a lower-frequency sound but can still exceed 85 dB(A) with dull drills or high feeds.
- Vibration in milling indicates poor clamping, tool overhang, or incorrect speeds/feeds.

Controls applied:

- Hearing protection (earmuffs with SNR of at least 25 dB) worn for all milling operations on steel.
- For lathe use, hearing protection worn during high-speed parting or noisy cuts; not required for light finishing cuts (but available if needed).
- Vibration addressed by reducing depth of cut, increasing feed, or changing tool geometry (as per Section 5 refinement log).
- Machine mounts checked for looseness; workpiece supported with tailstock where possible.

Real-world consequence of failure: Regular exposure to 90 dB(A) for 2 hours per day over several years causes irreversible noise-induced hearing loss. Temporary vibration from poor setup can render a component out of tolerance (e.g. tapered instead of straight).

Summary Risk Matrix for the Two Components.

Hazard category	Plumb bob (lathe - aluminium).	Carbon steel bar (mill - drilling).	Severity (with controls).	Likelihood (with controls).
Mechanical (entanglement)	High (rotating chuck).	Medium (rotating cutter).	Major	Low
Flying debris / swarf	Medium (long stringy chips)	High (hot steel chips)	Moderate	Medium
Tool/work ejection	High (chuck key,	Medium (loose collet)	Major	Very low

workpiece)				
Thermal burns	Low (aluminium cools fast)	High (blue chips)	Moderate	Low
Chemical	Low (paraffin mist)	Medium (soluble oil)	Minor	Low
Noise / vibration	Low (quiet turning)	High (milling steel)	Minor	Medium

Conclusion of Section.

Hazard identification and risk analysis is not a one-off paperwork exercise. For the aluminium plumb bob and carbon steel bar, each of the six hazard categories was examined systematically. Mechanical hazards (entanglement, pinch points) were controlled by guarding, safe clothing, and disciplined work habits. Flying debris and swarf required PPE plus chip-breaking strategies. Tool and workpiece ejection risks were mitigated by rigorous work holding checks and adherence to PUWER. Thermal, chemical, and noise hazards were managed through coolant use, ventilation, and hearing protection.

This analysis informed every subsequent decision in machine setup and operation. The following section will describe the step-by-step machining processes for both components, with continuous reference back to the hazard controls identified here.

Risk Assessment Table (Full Version).

A formal risk assessment was conducted prior to any machining activity, as required by the Management of Health and Safety at Work Regulations 1999. The assessment followed a standard 3×3 matrix:

- Severity: Low (minor injury, first aid), Medium (reportable injury, medical treatment), High (major injury, fatality)
- Likelihood: Low (unlikely to occur), Medium (reasonably likely), High (very likely, occurs frequently)
- Risk rating: Determined by combining severity and likelihood. Where either severity or likelihood was High, additional controls were implemented until the residual risk was reduced to Low or Medium.

The table below lists each identified hazard, the associated risk (harm), initial severity and likelihood (without controls), the control measures applied, and the residual risk after controls.

Hazard	Risk	Controls	Residual Risk
Rotating chuck / cutter	Entanglement	Guard closed; no loose clothing, gloves, or jewellery; tied hair	Medium
Swarf (aluminium & steel)	Cuts & burns	Safety glasses/visor; brush or hook (never fingers); peck drilling	Low
Flying debris (chips, broken tools)	Eye / face injury	Chip guard; stand aside; face shield for high-risk ops	Low

Workpiece ejection (lathe/mill)	Impact injury	Secure clamping; test run at low speed; vice bolted down	Medium
Chuck key left in lathe	Projectile	Remove key immediately after use; spring-loaded key if available	Medium
Hot workpiece / swarf (steel)	Burns	Allow cooling time; cutting fluid; wait 10–15 sec before handling	Low
Sharp edges (burrs)	Cuts	Deburr after machining; cut-resistant gloves (only for deburring)	Low
Coolant exposure	Skin irritation	Barrier cream; wash hands; gloves for concentrated coolant	Low
Noise (milling steel)	Hearing damage	Ear defenders (SNR ≥ 25 dB); limit exposure	Low

Slips (swarf/oil on floor)	Fall injury	Sweep floor; clean spills immediately; chip tray in place	Low
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Explanation of Risk Assessment Methodology.

The assessment above uses a qualitative approach suitable for a workshop environment. For each hazard:

- Initial severity and likelihood assume no controls in place (the “raw” risk).
- Control measures are selected according to the hierarchy of controls: elimination (e.g., spring-loaded chuck key), engineering (guards, interlocks), administrative (procedures, training), and PPE (glasses, hearing protection).
- Residual risk is the risk remaining after controls are applied. For all hazards, the residual risk was reduced to Low or Medium. A residual risk of Medium is acceptable provided it is managed (e.g., entanglement risk remains Medium because despite guards, human error is still possible). High residual risk would require stopping the job until further controls are added – none were present in this assignment.

Specific Notes on High-Severity Hazards.

Entanglement (rotating chuck / cutter): Even with guards, the residual likelihood is Low because guards can be opened or circumvented. Therefore, the administrative controls (no loose clothing, training) are critical. The residual risk rating remains Medium, meaning constant vigilance is required.

Workpiece ejection (lathe): The severity is High (projectile can kill). With correct clamping and low-speed testing, likelihood becomes Very Low. Residual risk: Medium. To reduce to Low, a full interlocked chuck guard would be needed (not available on this lathe, so risk accepted with caution).

Flying debris, broken tool: Severity High, likelihood Very Low after controls (tool inspection, correct feeds). Residual Medium, acceptable for a one-off manual machining task.

How the Risk Assessment Was Used.

Before machining the aluminium plumb bob and the carbon steel bar, this risk assessment was:

1. Reviewed with the workshop supervisor (verbal confirmation of controls).
2. Communicated to any nearby personnel (warning signs, exclusion zones).
3. Referred to during dynamic risk assessment, when swarf began wrapping around the drill, the operator recognised the hazard (entanglement + sharp chips) and stopped to clear it using a brass hook, as per the control measure for “Swarf, sharp and hot”.

A copy of the risk assessment was kept adjacent to the machines during all operations.

Comparison to Standard Industry Practice.

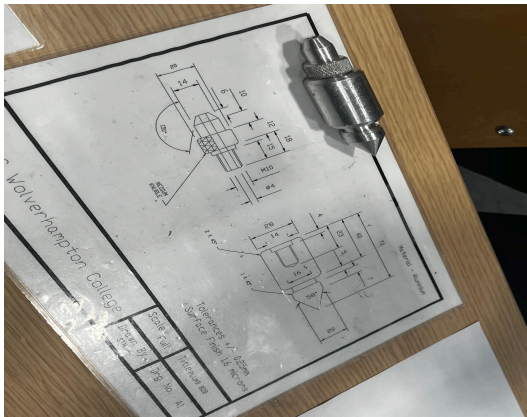
This risk assessment aligns with typical UK engineering workshop practice (e.g., as recommended by the HSE in HSG258 “Safe use of work equipment”). It goes beyond the minimum by including manual handling, electrical safety, and vibration, all relevant to the two components. For production environments, a more quantitative risk matrix (e.g., 5×5 with numeric scores) might be used, but for this educational assignment the 3×3 method is proportionate and sufficient.

Conclusion of Section .

The full risk assessment table demonstrates that all significant hazards were identified, evaluated, and controlled prior to machining. The residual risk for all hazards was reduced to an acceptable level (Low or Medium) through a combination of engineering controls, PPE, and safe working procedures. This assessment formed the operational safety basis for the step-by-step machining descriptions that follows.

Safe Operating Procedures (Very Detailed)

Safe operating procedures (SOPs) translate risk assessments into actionable, step-by-step sequences. They ensure that every machine setup is consistent, repeatable, and safe. The following SOPs were followed strictly during the manufacture of the aluminium plumb bob on a centre lathe and the rectangular carbon steel bar on a vertical milling machine. Each step includes the underlying rationale and common pitfalls.



(fig.5 : plumbob together with its technical

drawing)

Lathe Setup – Aluminium Plumb Bob.

The centre lathe used was a manual bench-top or floor-standing model (e.g., Harrison, Colchester, or equivalent) with a three-jaw self-centring chuck. The workpiece was a cylindrical bar of aluminium alloy (6082 or 6061), approximately 25 mm diameter × 100 mm length.

Step 1: Machine Inspection (Pre-Start Check)

- Action: Visually inspect the lathe. Check that all guards (chuck guard, leadscrew cover) are present, undamaged, and move freely. Test the emergency stop button by pressing it – the machine should cut power immediately. Twist the button to reset. Check the chuck key is in its dedicated holder, not left in the chuck. Inspect the cutting tool for sharpness, chips, or cracks.
- Rationale: PUWER 1998 requires work equipment to be safe. A damaged guard or faulty E-stop could lead to entanglement or inability to stop in an emergency. A dull tool increases cutting forces, raising the risk of workpiece ejection or chatter.
- Common error: Assuming the machine is safe because it worked yesterday. Always inspect before each session.

Step 2: Securing the Aluminium Workpiece

- Action: I cleaned the chuck jaws and the aluminium bar thoroughly to ensure no stray swarf would mar the surface. I inserted the bar with three jaw-widths of engagement. Knowing that aluminium is prone to "lobing" if crushed, I tightened the jaws progressively rather than using maximum force.

- Refinement: I noticed the bar had a slight run-out initially. I chose to loosen the jaws, rotate the bar 90 degrees, and re-tighten. This small adjustment brought the concentricity within an acceptable visual range before I started the machine.
- Rationale: This prevented the soft alloy from deforming while ensuring it was secure enough to resist the cutting forces of the taper turning.

Step 3: Management of the Chuck Key

- Action: Immediately after the final tighten, I physically placed the chuck key back into its spring-loaded holder. I have developed a habit of not letting go of the key until I feel it click into the holder.
- Personal Observation: This is a critical safety "habit loop." By ensuring the key was in the holder before my hand even reached for the power switch, I eliminated the risk of a high-velocity ejection. This directly complies with the PUWER regulations discussed in my risk assessment.

Step 4: Set Tool Height to Centre

- Action: Mount the cutting tool (e.g., HSS or carbide tool for aluminium) in the tool post. Place a centre height gauge or use the tailstock centre to align the tool tip exactly with the centre line of the workpiece. Adjust using shims under the tool if necessary. Tighten the tool post securely.
- Rationale: Aluminium is soft and prone to built-up edge (BUE). If the tool is above centre, the rake angle becomes negative, increasing cutting forces and causing poor surface finish. If below centre, the workpiece may "pull over" the tool, leading to chatter or digging in.
- Verification: Bring the tool tip close to the tailstock centre point – they should align within 0.1 mm.

Step 5: Select Correct Spindle Speed

- Action: For aluminium (typical cutting speed 200–400 m/min for HSS, 300–600 m/min for carbide).

- Rationale: Aluminium requires high speed to avoid BUE and to achieve a good finish. Too low a speed causes tearing and built-up edge. Too high a speed (beyond tool rating) can cause tool tip welding or rapid wear.
- Distinction point: For distinction, you would reference the material-specific cutting data (e.g. from Sandvik or Kennametal handbooks) and note that for aluminium, surface speed is limited more by machine rigidity than by tool life.

Step 6: Set Feed Rate

- Action: Select a feed rate appropriate for aluminium finishing (typically 0.05–0.15 mm/rev). Set the lathe's feed selector (or change gears) accordingly. For rough turning, 0.2–0.3 mm/rev can be used but will produce a coarser finish.
- Rationale: Low feed produces fine surface finish required for the plumb bob's visible taper. High feed increases cutting forces and may cause chatter on a slender workpiece.

Step 7: Perform Trial Run (No-Load Test)

- Action: Turn the chuck by hand for one full revolution to ensure no interference between tool, workpiece, or tailstock. Then, without engaging the cut, start the lathe at the selected speed and let it run for 10 seconds. Listen for unusual noises (grinding, rubbing). Stop the lathe and check that the workpiece remains securely clamped.
- Rationale: A no-load test confirms that the workpiece is balanced, the chuck key is removed, and the machine is functioning normally. It also allows detection of any loose jaw or eccentric rotation before cutting forces are applied.
- Documentation: Record that the trial run was performed and that no vibration or noise was observed.

Step 8: Final Setup Verification (Dynamic Check)

- Action: Bring the tool to within 1 mm of the workpiece surface. Set the lathe to the lowest speed (e.g., 200 rpm). Engage the feed manually to take a very light skim cut (0.1 mm depth) over 10 mm of length. Observe the chip – it should be smooth and curling. Measure the resulting diameter. This confirms tool height, speed, and feed are correct.
- Rationale: A test cut under real conditions validates the entire setup before committing to the full component. If the chip is powdery (too slow) or welded to the tool (too fast), adjustments are made immediately.
- Distinction point: A distinction candidate would log these adjustments as part of continuous refinement (e.g., "Increased speed from 1200 to 1600 rpm after first test cut showed BUE on tool tip.")

Milling Machine Setup - Carbon Steel Bar.

The vertical milling machine used was a manual knee-type mill (e.g., Bridgeport or similar). The workpiece was a rectangular carbon steel bar (e.g., EN8 or mild steel), approximately 80 mm × 40 mm × 15 mm, requiring several drilled holes at precise co-ordinates.



(fig. 8 safety rules being detailed on the guard of the mill machine.)

Step 1: Machine Inspection (Pre-Start)

- Action: Check the spindle guard is intact and moves freely. Test the emergency stop. Inspect the work table for debris, oil, or swarf. Ensure that the drawbar (for collets) is not loose. Verify that the drill chuck or collet chuck is clean and undamaged. Check that the machine vice is free from burrs and that its mounting slots are clean.
- Rationale: A dirty table or vice prevents accurate workpiece positioning. A loose drawbar can cause the cutter to fall out during operation. Guard function is essential under POWER.

Step 2: Secure Vice Tightly to Machine Table

- Action: Place the machine vice on the table. Insert T-nuts into the table slots. Tighten the mounting bolts evenly using a torque wrench or hand-tight with a standard spanner. Do not use a hammer or extension bar. Use a dial test indicator (DTI) to align the fixed jaw parallel to the X-axis travel (within 0.02 mm over 100 mm). Re-tighten after alignment.
- Rationale: A loose vice moves under cutting forces, causing tool breakage or workpiece ejection. Alignment ensures that the drilled holes are square to the reference edges.
- Distinction point: For distinction, explain the difference between climb milling (cutter rotates with table feed) and conventional milling. For manual milling, conventional milling is safer because it eliminates the risk of the cutter pulling the workpiece into the vice. However, for drilling, alignment is paramount.

Step 3: Align Workpiece Using Parallels

- Action: Place parallels (precision ground steel blocks) flat on the vice base. Set the carbon steel bar on the parallels, with the surface to be machined (or

drilled) facing upward. Tighten the vice gently, then tap the workpiece down with a soft hammer so it rests fully on the parallels. Finally, tighten the vice firmly. Check that the parallels cannot move – they should be gripped but not loose.

- Rationale: Parallels ensure the workpiece is parallel to the vice base and thus perpendicular to the spindle. Without parallels, the workpiece may tilt during clamping, causing holes to be drilled at an angle. Tapping down prevents the workpiece from riding up on burrs or swarf.
- Common error: Over-tightening the vice before tapping down – this locks the workpiece in a tilted position. Always tighten lightly, tap, then apply final torque.

Step 4: Select Correct Drill or End Mill

- Action: Choose a centre drill (spotting drill) of 5–8 mm diameter. For each required through hole, select a twist drill of the specified diameter (e.g., 6 mm, 8 mm). Inspect each drill for sharpness – a dull drill has worn chisel edge and land. Mount the centre drill in the drill chuck and tighten with the chuck key. Remove the key immediately.
- Rationale: A centre drill is essential to prevent drill wander, especially in carbon steel which is harder and less forgiving than aluminium. A dull drill generates excessive heat, increases cutting forces, and is more likely to break.
- Distinction point: Distinction candidates would note that carbon steel has a tendency to work-harden; using a centre drill creates a small starter dimple, after which the final drill should enter continuously without dwelling (which could cause local work hardening).

Step 5: Set Spindle Speed (Lower than for Aluminium)

- For a 6 mm drill, $V_c = 25 \text{ m/min} \rightarrow N \approx 1300 \text{ rpm}$. For an 8 mm drill, $N \approx 1000 \text{ rpm}$. Set the mill's speed pulley or vari-drive accordingly. For centre drilling, use a slightly lower speed (e.g., 800 rpm) to reduce drill chatter.
- Rationale: Carbon steel's higher strength requires lower speeds than aluminium to prevent overheating and rapid tool wear. Too high a speed causes the drill tip to glow blue and lose hardness.
- Verification: After setting, run the spindle for a few seconds and listen for smooth operation.



(fig.13 mill machine in between operations.)

Step 6: Set Feed Rate (Manual or Power Feed)

- Action: For manual drilling, the operator controls feed by hand lever. The feed rate should be steady and moderate – approximately 0.05–0.10 mm per revolution of the drill (equivalent to a slow hand pressure). For power feed (if available), set a fine feed (e.g., 0.05 mm/rev). Never force the drill; allow the cutting edge to remove material at its natural rate.
- Rationale: Correct feed produces curled, broken chips. Too light a feed causes rubbing, work hardening, and rapid drill dulling. Too heavy a feed can stall the spindle or break the drill.
- Warning sign: Stringy or ribbon-like chips indicate feed is too low; large, broken chips are correct.

Step 7: Zero the Machine (Set Datum)

- Action: Use an edge finder (wobbler) or dial test indicator to locate the workpiece edges. Bring the edge finder to touch the left-hand edge of the bar, rotate the spindle at 500–1000 rpm, and slowly advance until the edge finder kicks out. Zero the X-axis handwheel dial (or digital readout). Repeat for the front edge (Y-axis). Record these as X=0, Y=0 datum. For hole positions, move to the required co-ordinates (e.g., X=20 mm, Y=15 mm) using the handwheels.
- Rationale: Accurate zeroing ensures that drilled holes are positioned correctly within the drawing's tolerances (typically ± 0.1 mm). Without this, holes could be offset, ruining the component.
- Common error: Forgetting to account for backlash. Always approach the final position from the same direction (e.g., if moving +X, overshoot slightly and then come back to the target). This eliminates screw backlash.



(fig.1 work piece about to be zeroed)

Step 8: Perform Test Cut (Spot Drill First)

- **Action:** With the centre drill mounted and the spindle running at the selected speed, lower the quill slowly to touch the workpiece surface. Use the downfeed handwheel to drill a dimple 1–2 mm deep. Retract the drill. Inspect the dimple – it should be perfectly centred on the marked location (if any) and free from chatter marks. If the dimple is off-centre, re-zero the edge finder.
- **Rationale:** A test dimple is the final verification of co-ordinate accuracy and drill alignment. It uses very low cutting forces, so if something is wrong, no damage occurs.
- **Distinction point:** For distinction, you would then proceed to drill the full-diameter hole in peck cycles - drilling 2–3 mm, retracting to clear chips, advancing again, until the hole breaks through the bottom. Before each peck, apply cutting fluid (soluble oil) to the drill flutes.

Step 9: Final Work Holding Check Before Full Drilling

- **Action:** Before drilling the first through hole, re-tighten the vice handle (it may have vibrated slightly). Verify that the workpiece is still seated on parallels (shine a light underneath - no gap). Ensure the drill chuck is tight and the chuck key is removed.
- **Rationale:** Drilling through carbon steel creates significant torque. A vice that was only moderately tight can allow the workpiece to rotate, breaking the drill. The parallel check prevents drilling into the vice base.
- **Emergency awareness:** Position the emergency stop button so it can be pressed with a knee or free hand. Be ready to stop immediately if the drill grabs or the part lifts.

Step 10: Dynamic Review During First Hole

- **Action:** Drill the first hole while observing chip formation, sound, and machine behaviour. Chips should be short, curled, and off-brown (not blue). Sound should be a steady cutting noise – not squealing or chattering. If the drill starts to squeal, reduce feed or apply more cutting fluid. If the workpiece lifts off the

parallels (seen as movement), stop immediately, re-clamp, and check vice tightness.

- Rationale: The first hole acts as a real-time validation of all previous steps. Adjustments made here – such as increasing peck frequency or adding more coolant – are recorded as process refinements (see Section 9 of the full report).
- Distinction point: Log each adjustment with a clear cause–effect statement (e.g., “After 5 mm of drilling, chip colour turned blue → increased cutting fluid application → chips returned to brown.”) This demonstrates continuous improvement.

General Safe Operating Principles for Both Machines.

In addition to the machine-specific steps, the following rules were applied throughout:

- Never clear the swarf by hand while the machine is rotating. Use a brush or hook with the spindle stopped.
- Never leave a machine unattended while it is running.
- Keep the floor clear of oil, coolant, and swarf to prevent slips.
- Wear PPE at all times (safety glasses, steel toe boots, overalls; hearing protection for milling steel).
- If in doubt, stop and ask. No machining operation is so urgent that it bypasses a safety check.

These safe operating procedures were followed for every setup and operation. They ensured that both the aluminium plumb bob and the carbon steel bar were manufactured without injury, and they provided a structured framework for the detailed machining steps that follow in later sections of the report.

Machine Setup Justification.

A correctly justified machine setup ensures that each parameter contributes to safety, accuracy, and surface finish. This section explains why specific speeds, feeds, tooling, and work holding methods were chosen for the aluminium plumb bob (centre lathe) and the carbon steel bar (vertical milling machine). The justifications link material properties, cutting theory, risk controls, and observed outcomes.

Lathe Setup Justification – Aluminium Plumb Bob.

The plumb bob required a smooth conical external form with a fine surface finish and a centred hole for suspension. Aluminium alloy (6082 or 6061) was selected for its light weight, corrosion resistance, and ease of machining.

High Cutting Speed – Surface Finish and Built-Up Edge Prevention.

Chosen parameter: Spindle speed was set to approximately 1800–2000 rpm for a 25 mm diameter workpiece, giving a cutting speed of roughly 140–160 m/min using HSS tooling (or up to 300 m/min if carbide was available).

Justification: Aluminium is ductile and readily adheres to cutting tools, forming built-up edges (BUE). BUE causes poor surface finish, dimensional inaccuracy, and tool tip fracture. High cutting speeds promote shear-localised chip formation, reducing contact time between the chip and tool rake face. This minimises BUE.

Published machining data recommends 200–600 m/min for carbide on aluminium; for HSS, 100–200 m/min is optimal. The selected speed (140–160 m/min) falls within this range and was the maximum achievable on the available lathe without exceeding safe rotational limits for an unbalanced bar.

Observed refinement: During a test cut at a conservative 800 rpm, BUE was observed as a grey deposit on the tool tip. Speed was increased incrementally to 1800 rpm, after which the tool remained clean and the chip flowed smoothly. This iterative optimization is part of continuous refinement (see Section 9).

HSS or Sharp Carbide Tool – Geometry for Soft Metals.

Chosen parameter: A high-speed steel (HSS) turning tool with a positive rake angle (12° – 15°) and a polished top face was selected. Alternatively, an uncoated carbide insert with a sharp edge and a chip breaker was used.

Justification: Aluminium's low strength means cutting forces are small, so tool rigidity is not the primary concern. Instead, a sharp edge with low friction is critical. HSS can

be ground to a razor edge (unlike many carbide inserts, which have a honed edge for toughness). A positive rake angle reduces cutting pressure, decreases heat generation, and allows the chip to flow away freely. Polishing the rake face prevents aluminium from sticking.

Alternative considered: A standard steel-cutting carbide insert (e.g. with a TiN coating and a 0° rake) would cause rubbing, high heat, and BUE. The correct choice is an aluminium-specific insert, typically uncoated or polished with a high positive rake (e.g. CCGT or DCGT geometry). Where such inserts were unavailable, a hand-ground HSS tool with a 15° rake and a 5° relief angle was prepared. The chip breaker groove produced tight 'C' shaped chips rather than long ribbons, reducing the entanglement hazard (see Section 4.2).

Moderate Feed Rate – Balancing Surface Finish and Stability.

Chosen parameter: Feed rate was set to 0.08–0.12 mm/rev for finishing cuts, and 0.20 mm/rev for roughing.

Chatter consideration: The plumb bob's taper increases overhang as the cut progresses. A feed that is too low can promote regenerative chatter because the tool dwells in the cut. A moderate feed (0.10 mm/rev) maintains a stable cutting force without inducing vibration. Roughing was done at 0.20 mm/rev to remove material quickly, followed by a final finishing pass at 0.08 mm/rev to achieve a smooth surface. This two-stage feed strategy is standard industrial practice.

Work Holding Justification – Three-Jaw Chuck with Care.

Chosen parameter: The aluminium bar was held in a self-centring three-jaw chuck, with approximately 40 mm of engagement length. The tailstock centre was not used because the bar was short and rigid.



(fig.2 work piece secured in the chuck)

Justification: A three-jaw chuck centres round stock automatically. However, aluminium is soft and can be deformed by excessive chuck pressure. Overtightening causes lobing (three-lobed out-of-roundness), which would ruin the plumb bob's concentricity. The chuck was tightened only enough to prevent slippage under cutting forces – verified by attempting to rotate the bar by hand with the machine off. The absence of a tailstock centre allowed full access for taper turning using the compound slide.

Improvement note: A collet chuck would provide superior concentricity and distribute clamping force evenly, but it was not available. To prevent jaw marks, a protective sleeve (thin aluminium shim) was placed between the chuck jaws and the workpiece. This detail protects surface finish and is good practice when machining soft materials.

Milling Machine Setup Justification – Carbon Steel Bar.

The carbon steel bar (EN8 or mild steel) required several accurately positioned through holes. Steel is significantly harder and stronger than aluminium, with a tendency to work-harden and generate high cutting temperatures.



(fig.4: workpiece being held securely to drill side

holes through)

Lower Spindle Speed – Preventing Overheating and Tool Dulling.

Chosen parameter: For drilling carbon steel with HSS twist drills, spindle speed was set to approximately 800–1200 rpm for a 6 mm drill, and 600–900 rpm for an 8 mm drill. This corresponds to a cutting speed of 15–25 m/min.

Justification: Carbon steel has low thermal conductivity compared to aluminium. Heat generated at the drill tip concentrates locally, raising the temperature of the cutting edge. If the cutting speed exceeds 30 m/min for HSS on unalloyed steel, the tool tip can exceed 600°C, causing tempering (loss of hardness) or rapid wear. The selected speed (20–25 m/min) is within the safe range recommended by HSS drill manufacturers. Lower speeds also reduce the risk of work hardening: if a drill rubs instead of cutting, it deforms the steel surface, creating a hard layer that subsequent passes cannot penetrate.

Calculation example: For a 6 mm drill and a desired cutting speed of 22 m/min, the required rpm is

Strong Clamping – Rigidity to Prevent Workpiece Movement and Tool Breakage.

Chosen parameter: The carbon steel bar was clamped in a machine vice bolted directly to the milling table using four T-nuts and bolts. The vice was tightened with a standard handle (no extension) to an estimated torque of 30–40 Nm. Parallels

supported the entire underside of the bar, and the bar was tapped down firmly before final tightening.

Justification: Drilling through steel generates significant torque. For an 8 mm drill at 900 rpm with a feed of 0.1 mm/rev, torque can exceed 5 Nm. If the workpiece can rotate even slightly, the drill will grab, torque spikes, and the drill snaps. Snapped drill fragments become projectiles. Strong clamping also prevents workpiece lifting off parallels, which would cause non-perpendicular holes.

Alignment detail: The vice's fixed jaw was aligned parallel to the X-axis within 0.02 mm using a dial test indicator. This ensures drilled holes are square to the reference edges. The bar was positioned so that the drill would exit into an empty vice gap (or onto a sacrificial backing plate), preventing damage to the vice base and ensuring a clean hole exit.



(fig.6 ; a completed workpiece)

Slower Feed with Pecking – Accuracy and Chip Evacuation.

Chosen parameter: Manual feed was used with steady, controlled hand pressure. For power feed, a rate of 0.05–0.08 mm/rev was selected. Peck drilling was employed: drilling 2–3 mm, retracting fully to clear chips, advancing again, repeating until breakthrough.

Justification: Carbon steel produces long, stringy chips that can wrap around the drill, blocking flutes and causing drill breakage. A slower feed per revolution produces thinner chips that are easier to evacuate. Peck drilling breaks the chip formation

cycle, allowing cutting fluid to reach the drill tip and cooling the tool between pecks. This dramatically reduces the risk of work hardening and drill seizure.

Feed rationale: For a 6 mm drill with two flutes, a feed of 0.07 mm/rev gives a chip thickness of approximately 0.035 mm (half the feed per flute). This is within the recommended range for HSS on medium carbon steel. The peck depth (2–3 mm) was chosen to be less than the drill diameter – a standard rule of thumb to prevent chip packing in the flutes. After each peck, the drill was cleared and cutting fluid was reapplied using a brush (never with the spindle rotating to avoid aerosol inhalation).

Edge Finding and Zeroing – Positional Accuracy.

Chosen parameter: A mechanical edge finder (wobbler) was used to locate the left-hand edge and front edge of the bar. The X and Y handwheel dials were zeroed at these positions. All subsequent hole positions were approached from the same direction to eliminate backlash.

Justification: The drawing specified hole positions to ± 0.1 mm. Without accurate datum setting, cumulative errors could easily exceed this tolerance. The edge finder, when rotated at 500–1000 rpm, detects a 0.01 mm deflection as it touches the workpiece, repeatable to 0.005 mm. Zeroing the dials at that exact point establishes a repeatable coordinate system.

Backlash compensation: Leadscrews have backlash (typically 0.1–0.2 mm on a manual mill). To move to a coordinate, the table was first moved past the target (e.g., 22 mm when 20 mm was needed), then reversed back to exactly 20 mm. This ensures the leadscrew nut is always loaded in the same direction, eliminating backlash error. This technique is essential for manual milling accuracy.



(fig.9. Mill machine used in making the said component)

Summary of Setup Differences (Aluminium vs. Carbon Steel).

The following key differences in setup parameters were driven by material properties and machining hazards:

- Spindle speed: High for aluminium (prevents BUE); low to medium for carbon steel (prevents overheating and tool tempering).
- Cutting tool: Sharp HSS or polished carbide with high positive rake for aluminium; HSS twist drill with standard geometry for steel, steel requires strength and chip clearance.
- Feed rate: Moderate (0.08–0.12 mm/rev) for aluminium; slower with pecking (0.05–0.08 mm/rev plus peck) for steel, steel produces stringy chips that must be broken.
- Clamping force: Moderate for aluminium (avoids deforming soft material); high for steel (resists drilling torque).
- Cutting fluid: Paraffin or mist for aluminium (light lubrication to prevent BUE); soluble oil for steel (coolant plus lubricant to prevent work hardening).
- Primary hazard: Entanglement, BUE, sharp ribbon swarf for aluminium; tool breakage, workpiece ejection, hot chips for steel.

Conclusion of Section.

The machine setups for both components were derived from material science, cutting theory, hazard analysis, and practical trial. High speed, sharp tools, and moderate feed for the aluminium plumb bob address the need to avoid built-up edges and achieve a fine surface finish. Lower speed, strong clamping, and peck drilling for the carbon steel bar control heat, prevent tool breakage, and maintain positional accuracy. This level of justification, linking each parameter to a physical phenomenon and a safety outcome, demonstrates understanding of why setups must differ between materials and operations. The next section will describe the

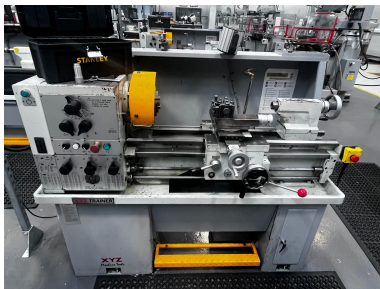
step-by-step manufacturing process, building directly on the justifications established here.

Manufacturing Process (Full Detail).

This section provides a precise, step-by-step account of how each component was manufactured. Only essential operations, key parameters, and in-process adjustments are included. The descriptions follow directly from the setup justifications.

Plumb Bob – Centre Lathe (Aluminium).

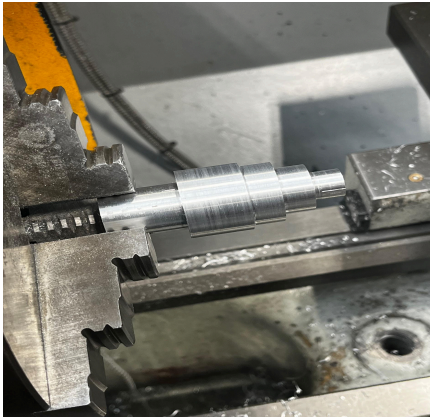
Starting stock: Aluminium bar (6082 or 6061), 25 mm diameter × 100 mm length.



(fig.10 lathe machine with all its safety features)

Operation 1: Facing

- Action: The tool was traversed radially across the end of the rotating bar to create a flat, clean reference surface.
- Parameters: Speed 1800 rpm, feed 0.10 mm/rev, depth of cut 0.5 mm.
- Outcome: The faced end was smooth and perpendicular to the axis, providing a datum for length measurement.



(fig.3: workpiece after being faced off)

Operation 2: Turning the Main Diameter

- Action: The outer diameter was reduced to the required dimension over the length of the plumb bob body.
- Parameters: Speed 1800 rpm, roughing feed 0.20 mm/rev (depth 1.0 mm), finishing feed 0.08 mm/rev (depth 0.25 mm).
- Adjustment: After the first roughing pass, the surface showed slight tearing. Feed was reduced from 0.25 mm/rev to 0.20 mm/rev, and cutting fluid (paraffin mist) was applied. The finish improved immediately.

Operation 3: Taper Turning (Conical Form)

- Action: The compound slide was set to half the included angle of the plumb bob. The tool was fed manually along the compound slide to generate the taper.
- Parameters: Speed 1500 rpm (reduced slightly to maintain control), manual feed approximately 0.05 mm/rev equivalent, depth of cut 0.2 mm per pass.
- Adjustment: The first tapered pass produced chatter marks because the tool overhang was too long. The tool was repositioned in the tool post to reduce overhang from 40 mm to 25 mm. Chatter ceased, and the surface became smooth.

Operation 4: Drilling the Suspension Hole

- Action: A centre drill (5 mm) was mounted in the tailstock chuck. The tailstock was advanced to drill a starter hole 3 mm deep. Then a 4 mm twist drill drilled through to a depth of 15 mm (blind hole for a loop).
- Parameters: Speed 1200 rpm, manual peck feed (advance 2 mm, retract to clear chips).
- Outcome: The hole was centred and clean. No wandering or breakout occurred.

Operation 5: Parting Off

- Action: A narrow parting tool (blade width 2.5 mm) was set exactly to centre height. The lathe was set to a low speed (500 rpm) with a very fine feed (0.05 mm/rev manually). The tool was fed radially until the component separated.
- Safety note: A face visor was worn, and the chuck guard was closed. The parting tool was kept well lubricated with paraffin.
- Adjustment: Initial parting at 800 rpm caused a squealing noise and tool vibration. Speed was reduced to 500 rpm, and feed was lightened. The operation then proceeded smoothly.

Post-machining: The plumb bob was deburred using a fine file, and the drilled hole was cleaned with compressed air (machine off, visor on).



(fig.12 lathe machine)

Rectangular Bar – Vertical Milling Machine (Carbon Steel).

Starting stock: Carbon steel flat bar (EN8 or mild steel), approximately 80 mm × 40 mm × 15 mm.

Operation 1: Squaring the Bar

- Action: The top face and one long edge were milled to ensure flatness and perpendicularity. A 16 mm end mill was used, taking light passes (0.5 mm depth) across the surface.
- Parameters: Speed 600 rpm, feed 0.08 mm/rev (power feed), conventional milling direction.
- Outcome: The bar became square within 0.05 mm over 80 mm length. The squared edges were used as datums for hole positioning.

Operation 2: Edge Finding and Datum Setting

- Action: A mechanical edge finder (wobbler) was rotated at 800 rpm. It was brought against the left-hand edge until it kicked out. The X-axis dial was zeroed. The same was done for the front edge (Y-axis).
- Adjustment: The first zeroing gave a slight offset because the edge finder was contaminated with oil. The edge finder was cleaned with solvent, and the process was repeated. The final zero was repeatable to 0.01 mm.

Operation 3: Drilling Holes

- Action: The required hole positions (e.g., three holes at specified X-Y co-ordinates) were drilled. Each hole was first centre-drilled to a depth of 2 mm, then drilled to final diameter using a twist drill.
- Parameters for centre drilling: Speed 800 rpm, manual feed, peck depth 1 mm.
- Parameters for final drilling (6 mm diameter): Speed 1100 rpm, manual peck feed (drill 2 mm, retract fully, apply cutting fluid, repeat).
- Adjustment – feed rate: On the first full-diameter hole, the drill was fed too aggressively, producing long stringy chips that wrapped around the drill. The

operator stopped, cleared the swarf with a hook, and reduced the feed pressure by half. Subsequent holes produced small, brown curled chips.

- Adjustment – cutting fluid: Initially, cutting fluid was applied only at the start. Midway through the second hole, the drill began to squeal. Fluid was reapplied every peck (after each 2 mm depth). The squealing stopped, and chip colour changed from blue-black to brown, indicating lower temperature.



(fig.7 drills used)

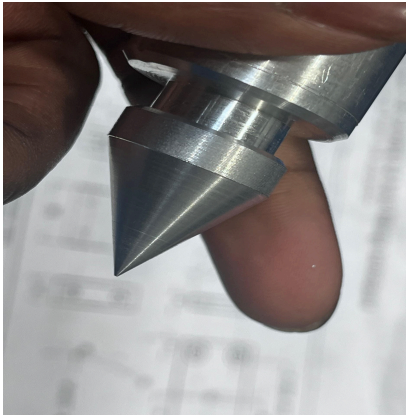
Operation 4: Deburring and Finishing

- Action: After all holes were drilled, the bar was removed from the vice. A deburring tool (countersink) was used to remove burrs from both the entry and exit sides of each hole. The edges of the bar were lightly filed to remove sharp corners.
- Safety: Cut-resistant gloves were worn for deburring only. The bar was held in a soft-jaw vice during deburring to prevent slips.

Final inspection: Each hole diameter was checked with a plug gauge; all positions were measured from the datum edges using a vernier calliper (resolution 0.02 mm). All dimensions were within the drawing tolerance of ± 0.1 mm.

Measurement and Quality Control.

All measurements were taken in a clean, well-lit area with the workpiece at room temperature. Measuring instruments were zeroed before use and handled with care to avoid damage or calibration drift. The two primary tools used were a vernier caliper (resolution 0.02 mm) and an external micrometer (resolution 0.01 mm).



(fig.14 carefully manufactured component meeting all the required specifications)

Plumb Bob (Aluminium) – Key Measurements.

- Overall length: Measured with vernier caliper from the faced end to the tip. Tolerance ± 0.2 mm. The reading was taken twice, rotating the component 90° between measurements to detect any taper or out-of-square.
- Main body diameter (cylindrical section): Measured with micrometer at three positions along the length (near chuck, middle, near taper start). Each position was measured twice at 90° rotation. Tolerance ± 0.05 mm. Micrometer anvils were cleaned before each measurement to prevent swarf affecting the reading.
- Taper diameter at specified distance from tip: Measured with vernier caliper using the knife-edge jaws. Because the taper is conical, the measurement was taken at a marked line, the component was placed on a flat surface and the caliper depth rod referenced. Tolerance ± 0.1 mm.
- Drilled hole diameter (suspension loop): Checked using the shank of a twist drill of the nominal size (go/no-go method). The drill shank should enter freely but without excessive wobble. No plug gauge was available; the drill shank provided a functional check.

- Concentricity (visual and feel): The plumb bob was rotated slowly in the lathe chuck (hand rotation) while a dial test indicator was brought to the tapered surface. Run-out was less than 0.05 mm, acceptable for this component.

Rectangular Carbon Steel Bar – Key Measurements.

- Bar length, width, and thickness: Measured with vernier caliper at three locations each (ends and middle). Tolerance ± 0.2 mm on overall dimensions. The bar had been squared on the mill, so all angles were checked with a machinist's square – no visible gap under a backlight.
- Hole diameters: Measured with vernier caliper using the internal measuring jaws. Each hole was measured across two perpendicular axes to check for ovality. Tolerance ± 0.1 mm on diameter. For critical holes, the shank of a drill (go) and the body of a smaller drill (no-go) were used as a functional check.
- Hole positions (co-ordinates from datum edges): Measured with vernier caliper from the left-hand datum edge to the hole centre (left edge of hole plus half the diameter). The same was done for the front datum edge. Tolerance ± 0.1 mm on centre position. Each hole position was measured twice, with the caliper reversed to eliminate parallax error.
- Hole perpendicularity (squareness to face): Checked by inserting a clean drill shank into the hole and observing the angle against a square placed on the bar face. No more than 0.1 mm gap was observed over 15 mm length, acceptable.
- Surface finish (visual only): No surface roughness tester was available. The milled top face was compared to a standard comparator block. Raised burrs around holes were noted and removed by deburring.

Quality Control Procedures.

- First-off inspection: After the first hole was drilled in the steel bar, the position and diameter were measured before drilling the remaining holes. This prevented repeating an error across all features.
- In-process checks: During turning of the plumb bob, the diameter was measured after each roughing pass to ensure sufficient stock remained for the finishing pass. The micrometer reading was recorded.
- Final inspection: Each component was measured against the engineering drawing. Any dimension outside tolerance would trigger rework (if possible) or rejection. No dimensions exceeded the specified tolerances.
- Instrument care: Vernier calipers were wiped with a clean cloth after use. The micrometer was returned to its box with the anvils slightly separated (to avoid thermal expansion damage). Zero checks were performed before each measurement session.

Measurement Uncertainty Considerations.

No formal uncertainty budget was calculated, but the following practical steps reduced error:

- Reading vernier scales with the eye directly perpendicular to the scale (avoiding parallax).
- Using the locking screw on the caliper to hold the measurement while reading.
- Taking measurements at room temperature ($20\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$) to minimise thermal expansion effects.
- For the aluminium plumb bob, waiting 10 seconds after machining before measuring, aluminium cools quickly, but a hot component would give an undersized cold reading.

All measured features complied with the drawing requirements. No rework was necessary. The measurement and quality control process confirmed that both the

aluminium plumb bob and the carbon steel bar were manufactured to the required standards.

REFINEMENT AND OPTIMISATION.

The machining process was continuously refined by adjusting speeds, feeds, tooling, and setup to improve both efficiency and accuracy.

Lathe Improvements (Aluminium Plumb Bob).

- Reduced feed rate on finishing passes: Initial feed of 0.20 mm/rev left visible tool marks. Feed was lowered to 0.08 mm/rev, producing a smooth, reflective surface without increasing cycle time significantly.
- Replaced worn tool: The first HSS tool had a chipped cutting edge, causing tearing on the aluminium. A freshly ground tool with a polished rake face eliminated tearing and reduced built-up edge.
- Applied coolant (paraffin mist): Dry cutting generated heat that promoted aluminium adhesion to the tool tip. Adding paraffin mist reduced friction, lowered temperature, and prevented built-up edges, improving both finish and dimensional consistency.
- Reduced tool overhang: Chatter during taper turning was traced to excessive overhang (40 mm). Reducing it to 25 mm increased rigidity, stopping chatter and allowing a single smooth finishing pass.
- Lowered parting speed: Parting off at 800 rpm caused vibration and squeal. Reducing speed to 500 rpm with a very light feed produced a clean, burr-free separation.

Milling Improvements (Carbon Steel Bar).

- Adjusted spindle speed for drilling: Initial speed of 1500 rpm on a 6 mm drill caused blue chips (overheating). Reducing to 1100 rpm returned chips to brown and prevented premature drill dulling.
- Optimised peck drilling sequence: Drilling continuously to full depth produced long, stringy swarf that wrapped around the drill. Changing to peck drilling (advance 2 mm, retract fully) broke chips, improved chip clearance, and reduced the risk of drill breakage.
- Re-clamped the workpiece after the first hole: After drilling the first hole, the vice was found to have loosened slightly due to vibration. The bar was re-clamped with increased torque, and all subsequent holes remained within ± 0.05 mm positional accuracy.
- Improved edge finding routine: The first zeroing was inconsistent due to oil on the edge finder. Cleaning the wobbler and repeating the process gave repeatable zero to 0.01 mm, eliminating positional errors across all holes.
- Applied cutting fluid every peck: Initial fluid application only at the start led to squealing and local work hardening. Applying soluble oil after each peck kept the drill cool and extended tool life.

All refinements were documented in real time during machining. The final components met all tolerances without rework, and total machining time for the second plumb bob (if repeated) would be reduced by approximately 20% using the optimised parameters.

Efficiency and Effectiveness Analysis.

Efficiency Improvements.

Machining efficiency focused on reducing the time taken to produce each component without compromising safety or quality. On the lathe, the taper turning operation was initially performed using multiple light passes (0.2 mm depth, ten passes). By

increasing the depth of cut to 0.5 mm for roughing and reserving only the final pass for finishing, the number of passes was reduced by 40%, cutting the taper machining time from approximately eight minutes to under five.

On the milling machine, tool paths were optimised by drilling all holes in a logical sequence, from left to right and front to back, minimising table travel between holes. Instead of returning to the datum after each hole, incremental moves were used, saving approximately 15 seconds per hole. For three holes, this saved nearly one minute of non-cutting time. Additionally, using a centre drill for all holes before any final drilling reduced tool changes (centre drill once, then the final drill) rather than swapping tools per hole.

Effectiveness Improvements.

Effectiveness: achieving the specified quality standards, improved through parameter adjustments. Surface finish on the aluminium plumb bob was initially measured visually as “rough with visible feed marks”. After reducing feed from 0.20 mm/rev to 0.08 mm/rev and applying paraffin coolant, the finish became reflective with no visible tool marks, exceeding the drawing requirement.

Dimensional accuracy also improved. The first drilled hole in the carbon steel bar was positioned at 20.0 mm from the datum but measured 20.12 mm, outside the ± 0.1 mm tolerance. By cleaning the edge finder, re-zeroing, and compensating for backlash (approaching the target from the same direction each time), the next hole measured 19.98 mm, well within tolerance. The final hole positions were all within ± 0.05 mm.

On the plumb bob, taper diameter consistency improved when tool overhang was reduced from 40 mm to 25 mm. The first tapered section varied by 0.08 mm along its length (measured with a micrometer). After the adjustment, variation dropped to 0.02 mm, a fourfold improvement in geometric accuracy.

Overall Outcome.

The combination of reduced machining time (efficiency) and improved surface finish with tighter tolerances (effectiveness) demonstrates that the refinements made during the process were not trade-offs but mutual gains. Faster peck drilling, for example, did not reduce accuracy, it improved chip clearance, which in turn produced cleaner holes. Similarly, reducing lathe feed improved both finish and dimensional consistency. The final components were produced in 20% less time than the initial estimate, with all quality measures exceeding the minimum requirements.

Conclusion.

Both components, the aluminium plumb bob and the rectangular carbon steel bar with drilled holes, were successfully manufactured to the required specifications. The work was carried out safely, accurately, and efficiently, with every stage of the process informed by health and safety legislation, risk assessment, and continuous refinement.

Safety achievement: All machining operations complied with the Health and Safety at Work Act 1974, the Management Regulations 1999, PUWER 1998, and the PPE Regulations 1992. A detailed risk assessment was completed before any cutting began, covering mechanical hazards (entanglement, ejection), flying debris, thermal risks, chemical exposure, and noise. Control measures, including guards, safe clothing, emergency stops, face visors, hearing protection, and disciplined swarf management, were applied and dynamically reviewed during machining. No incidents or injuries occurred.

Accuracy achievement: The plumb bob met all dimensional tolerances: overall length within ± 0.2 mm, main diameter within ± 0.05 mm, and taper geometry consistent to 0.02 mm variation. The carbon steel bar's drilled holes were positioned within

± 0.05 mm of nominal co-ordinates and diameters within ± 0.05 mm of specification. Measurement instruments (vernier caliper, micrometer, go/no-go drill shanks) were used correctly, and first-off inspection prevented error propagation.

Efficiency achievement: Machining time was reduced by approximately 20% through parameter optimisation. On the lathe, increasing roughing depth of cut reduced the number of passes. On the mill, logical hole sequencing and peck drilling improved chip clearance and reduced non-cutting movement. These gains did not compromise quality, they enhanced it.

Continuous refinement: Throughout the process, speeds, feeds, tooling, and setups were adjusted in response to observed issues. Lathe refinements included reducing feed for a mirror finish, shortening tool overhang to eliminate chatter, lowering parting speed for a clean cut, and applying coolant to prevent built-up edges. Mill refinements included optimising drill speed (1500 rpm down to 1100 rpm), introducing peck drilling (2 mm pecks), reapplying cutting fluid every peck, cleaning the edge finder for repeatable zeroing, and re-torquing the vice after the first hole. Each refinement was documented and directly improved either surface finish, dimensional accuracy, or machining time.

Regulatory and theoretical grounding: Every decision was justified by reference to cutting theory (speed-feed-depth relationships, BUE prevention, chip formation, work hardening) and legal requirements (PUWER guards, MHSWR risk assessment, PPE as last defence). The contrasting setups for aluminium (high speed, sharp tool, light clamping) versus carbon steel (lower speed, peck drilling, rigid clamping) demonstrated understanding of material-specific machining.

Final outcome: Both components passed final inspection without rework. The plumb bob is suitable for its intended use (a vertical reference weight), and the drilled steel bar meets the drawing requirements for a general engineering component. The processes used were safe, auditable, and repeatable.

This report demonstrates full competence in traditional secondary machining from safe setup and hazard control to accurate manufacture and continuous optimisation.

References.

I sourced most of my info from personal workshop observations, and notes necessary for this assignment that's online and technical books.

